Multigraph Dual-Level Knowledge Distillation Integration for Few-Shot Dissolved Gas Analysis

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***Abstract*— Dissolved gas analysis (DGA) is a key technique for monitoring the insulation condition of oil-immersed power transformers, capable of identifying potential faults by analyzing the composition and concentration of dissolved gases in the oil. However, traditional DGA methods rely heavily on labeled samples, which limits their diagnostic performance when labeled data are scarce. To address this limitation, we propose the multigraph dual-level knowledge distillation integration (MDKDI) framework for few-shot DGA, an innovative approach that combines multigraph modeling with knowledge distillation to alleviate the constraints imposed by limited labels. Our approach begins by constructing graph structures to capture the complex relational patterns among samples. To enhance the representation utility of unlabeled data, we introduce feature and edge perturbation mechanisms to generate augmented views. Furthermore, we design a dual-level distillation architecture to enhance knowledge transfer efficiency. Within this framework, a primary teacher model guides the student model in feature refinement through a co-guided learning strategy, while an integrated advanced teacher model delivers augmented supervisory signals to further optimize the learning process. Experimental results on DGA2000 dataset show that MDKDI achieves an average 15.4% improvement in *F*1-score for few-shot diagnosis compared to state-of-the-art methods, demonstrating superior robustness and generalization capabilities.**

***Index Terms*— Dissolved gas analysis (DGA), few-shot diagnosis, knowledge distillation, multigraph modeling.**

I. Introduction

POWER transformers are essential for voltage conversion in power systems, enabling efficient long-distance transmission and safe end-user consumption [1]. Among them, oil-immersed transformers are widely used due to their excellent insulation and cooling performance. However, their complex internal structure—including core, windings, insulating oil, and cooling systems—makes them prone to faults such as insulation aging, oil degradation, partial discharge, and overheating, which can lead to equipment failure and power outages [2]. These issues often cause the accumulation of dissolved gases (e.g., H_2, CH_4, C_2H_2) in the transformer oil [3]. Dissolved gas analysis (DGA) is a key technique for diagnosing such faults by analyzing gas types and concentrations, providing vital information for preventive maintenance and enhancing the reliability and safety of power systems [4].

The diagnostic approach of DGA has evolved through several distinct phases. Initially, methods relied on thermodynamic principles and ratio coding rules [5], [6], [7], [8]. The advent of machine learning introduced algorithms like *K*-nearest neighbors (KNNs) to learn the relationship between gas composition and fault types [9], [10], [11]. Subsequent research developed advanced variants [12], [13], [14], [15], [16], [17], such as swarm-optimized SVM [12] and other intelligent, integrated systems [13], [14] to boost accuracy. More recently, the field has been transformed by deep learning [18]. Back propagation neural networks (BPNNs) improved feature abstraction [19], paving the way for more sophisticated physics-informed models that incorporate domain knowledge for greater robustness [20], [21]. In parallel, graph-based methods like graph neural networks (GNNs) emerged to capture topological relationships among samples [22], [23], with recent advances like dual-channel graph ranking contrastive learning further extending the potential of graph representation for DGA tasks [24].

Despite the significant progress achieved by existing methods in DGA fault diagnosis, most of them have primarily been designed for scenarios with sufficient labeled data. As a result, their performance under few-shot conditions remains limited, particularly in two key areas.

1) *Single-view modeling limitations:* Most current methods establish sample relationships from a single perspective, which inadequately captures the complex interdependencies among multidimensional features. This results in insufficient information representation, an issue that is especially critical in scenarios with limited samples and can significantly compromise the accuracy of diagnostic results.

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2) *Complexity of multiview fusion:* While multiview modeling can provide complementary information, the heterogeneity between different views (such as noise interference, feature redundancy, and semantic conflicts) poses greater challenges for information integration. The absence of robust fusion strategies can disrupt the model's learning process, thereby adversely affecting diagnostic performance.

To tackle the above challenges, this article introduces a novel few-shot DGA method termed the multigraph dual-level knowledge distillation integration (MDKDI) framework. This framework is built upon two pivotal modules: the multigraph construction module and the graph knowledge distillation module. Initially, MDKDI constructs both similarity and dissimilarity graphs from the original DGA features to holistically capture the inter-sample relationships. To further enrich the model's understanding, both feature perturbation and edge perturbation techniques are applied to the similarity graph, creating augmented views that collectively form a multigraph structure. This structure bolsters the model's ability to robustly represent complex sample relationship patterns. Subsequently, MDKDI employs a dual-level knowledge distillation framework to harness the full potential of unlabeled data within the multigraph learning context, thereby facilitating effective multiview information integration. The process begins with the student model extracting knowledge from the primary teacher model, leveraging insights from unlabeled nodes. A co-guided learning strategy is then implemented, wherein the primary teacher model and the student model work together to form an advanced teacher. This advanced teacher model offers enhanced supervisory signals to both the student and the primary teacher, leading to substantial enhancements in fault detection performance.

The key contributions of this article are outlined as follows.

1) We introduce an innovative multigraph dual-level knowledge distillation framework, which substantially improves the utilization efficiency of unlabeled data by comprehensively integrating DGA sample correlations.
2) A teacher–student co-evolution mechanism is devised to facilitate multilevel knowledge transfer, leveraging primary knowledge refinement and advanced integrated supervision for enhanced learning.
3) Extensive experimental results on two publicly available DGA datasets confirm the effectiveness of the MDKDI framework in few-shot fault diagnosis scenarios.

II. Preliminaries

Notations: We use bold uppercase letters (e.g., $\mathbf{X}$), bold lowercase letters (e.g., $\mathbf{h}$), and calligraphic fonts (e.g., $\mathcal{G}$) to denote matrices, vectors, and sets, respectively. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represent an undirected graph with nodes $|\mathcal{V}|$ and edges $|\mathcal{E}|$, with the edge connecting nodes v_i and v_j denoted as e_{ij}. The node in $\mathcal{G}$ is described by the DGA attribute matrix $\mathbf{X} \in \mathbb{R}^{n \times f}$, where f is the dimensionality of the node features. The structure of $\mathcal{G}$ is described by the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, where $\mathbf{A}_{ij} = 1$ if $e_{ij} \in \mathcal{E}$, and $\mathbf{A}_{ij} = 0$ otherwise. The symmetrically normalized adjacency matrix is denoted as $\hat{\mathbf{A}} = \mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$, where $\mathbf{D}$ is the degree matrix of $\mathbf{A}$. The Laplacian matrix of the graph is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and the symmetrically normalized Laplacian matrix is $\hat{\mathbf{L}} = \mathbf{I} - \hat{\mathbf{A}}$. The learned node embedding matrix is represented as $\mathbf{H}$.

Problem Definition: Given a DGA attribute matrix $\mathbf{X}$, a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed. $\mathcal{V}$ is then divided into two subsets: $\mathcal{V}^L$, consisting of nodes explicitly labeled with the label $\mathbf{Y}^L$, and $\mathcal{V}^U$, consisting of nodes that remain unlabeled ($|\mathcal{V}^L| \ll |\mathcal{V}^U|$). The goal of DGA fault diagnosis is to develop a model that can accurately predict the fault categories $\mathbf{Y}^U$ of DGA samples in $\mathcal{V}^U$ based on the labels of $\mathcal{V}^L$ and the features of all nodes in $\mathcal{V}$.

In our work, two information aggregation strategies are employed to update node representations within the constructed multigraph structure, respectively [23]. In instances where neighboring nodes exhibit similarity, an aggregation strategy that utilizes the adjacency matrix is implemented to enhance node representations. This methodology strengthens the similarity characteristics among nodes, consequently enriching the model's comprehension of local structures. Conversely, for interconnected nodes that demonstrate dissimilarity, the graph Laplacian matrix is employed for aggregation, aiming to emphasize node features. This process guarantees that feature distinctions are more pronounced throughout the learning phase.

III. Methodology

A. Overview

This section offers a comprehensive overview of the proposed MDKDI. The method introduces a multigraph construction module designed to thoroughly explore the diverse information within the data. To harness the potential of unlabeled samples, a dual-level knowledge distillation module is implemented, extracting comprehensive knowledge at both the initial and advanced stages. This approach ensures that the model fully captures the rich information within the multigraph structure while effectively guiding the unlabeled samples. A co-guided learning strategy is employed to enhance node embedding representations by integrating insights from the primary teacher model and the student model. This integration enables the development of a more advanced teacher model, providing more reliable guidance. The complete framework of the model is illustrated in Fig. 1.

B. Multigraph Construction

To deeply explore and fully utilize the information contained within a limited number of samples, two graph construction strategies are implemented: 1) *KNN similarity and dissimilarity graph construction:* This strategy aims to achieve a more comprehensive learning effect. The model constructed based on this strategy is denoted as MDKDI_1 and 2) *augmented graph views:* By applying random perturbations to the node features and edges of the similarity graph, two complementary augmented views are generated to enrich the learning content further. The model constructed based on this strategy is denoted as MDKDI_2.

1) MDKDI_1 Graph Construction: Both similarity graph ($\mathcal{G}^{(s)}$) and dissimilarity graph ($\mathcal{G}^{(t)}$) are constructed to deeply explore the complex relationships among samples [25].

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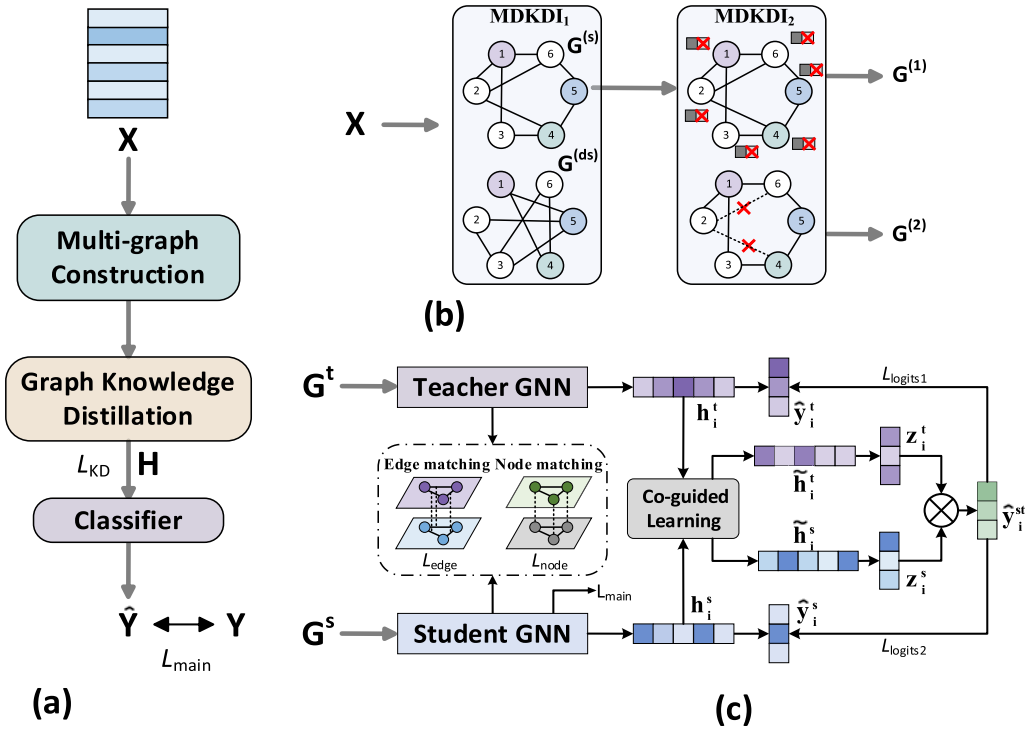


Fig. 1. Overall architecture of the MDKDI model. (a) Training process of MDKDI, where L_{KD} and L_{main} represent two different loss functions. (b) Multigraph construction module transforms the **X** into a multigraph structure via two strategies. (c) Multigraph structures generated in (b) are input into both the student GNN and the teacher GNN to execute the dual-level knowledge distillation process.

For any node v_i, its similarity with another node v_j can be formalized as follows:

$$s_{ij} = \frac{\mathbf{x}_i^{\mathrm{T}}\mathbf{x}_j}{\|\mathbf{x}_i\|\|\mathbf{x}_j\|} \tag{1}$$

where $s_{ij} \in \mathbf{S}$ denotes the similarity between nodes v_i and v_j, and $\mathbf{x}_i$ and $\mathbf{x}_j$ are the feature vectors of the ith and jth rows in the feature matrix $\mathbf{X}$, respectively.

For node v_i, based on the similarity matrix $\mathbf{S}$, its KNNs are selected, and the indices of these neighbors are added to the set $\mathcal{N}_i$. The adjacency matrix $\mathbf{A}^{(s)}$ of the similarity graph is constructed, with its elements defined as follows:

$$\mathbf{A}_{ij}^{(s)} = \begin{cases} 1, & \text{if } v_j \in \mathcal{N}_i \text{ or } v_i \in \mathcal{N}_j \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\mathcal{N}_i$ represents the neighbor set of node v_i, and $\mathcal{N}_j$ denotes the neighbor set of node v_j.

In the above similarity graph, adjacent nodes exhibit high similarity. A low-pass filter is introduced for feature aggregation to make the feature representations of similar nodes closer, which is formalized as follows:

$$\mathbf{H}_l^{(s)} = \sigma\left(\hat{\mathbf{A}}^{(s)}\mathbf{H}_{l-1}^{(s)}\mathbf{W}_{l-1}^{(s)}\right) \tag{3}$$

where $\mathbf{H}_{l-1}^{(s)}$ and $\mathbf{H}_l^{(s)}$ represent the embeddings at the $(l-1)$th and lth layers, respectively, $\mathbf{W}_{l-1}^{(s)}$ is a learnable parameter matrix, and σ is a nonlinear activation function.

Considering that single-view information may not fully capture the complex relationships between samples and that feature smoothing in GNNs on similarity graphs may lead to overfitting, a dissimilarity KNN graph is constructed to capture differences between samples. When selecting edges, the K samples with the lowest similarity to the current sample are connected, forming the adjacency matrix $\mathbf{A}^{(\mathrm{ds})}$ of the dissimilarity graph $\mathcal{G}^{(\mathrm{ds})}$. During the node feature update process, a high-pass filter is introduced for feature extraction, enabling connected dissimilar nodes to be effectively distinguished. This is formalized as follows:

$$\mathbf{H}_l^{(\mathrm{ds})} = \sigma\left(\hat{\mathbf{L}}^{(s)}\mathbf{H}_{l-1}^{(\mathrm{ds})}\mathbf{W}_{l-1}^{(\mathrm{ds})}\right) \tag{4}$$

where $\hat{\mathbf{L}} = \mathbf{I} - \hat{\mathbf{A}}^{(\mathrm{ds})}$ denotes the graph Laplacian matrix of the dissimilarity graph $\mathcal{G}^{(\mathrm{ds})}$.

2) MDKDI$_2$ Graph Construction: To bolster data diversity and strengthen the model's capacity to learn robust features, a random perturbation strategy is employed leveraging the similarity graph. In each augmented view, two types of perturbations are simultaneously applied: node feature perturbation, which randomly masks elements in the node feature matrix to enrich feature diversity and reduce overfitting, and edge perturbation, which randomly removes a subset of edges to slightly modify the graph topology and improve structural robustness. By jointly applying these perturbations to each augmented view, the framework generates two diversified graph instances, enabling the model to learn more robust and generalizable representations under few-shot conditions. Formally, the perturbed graphs are denoted as $f_\theta \to \mathcal{G}^{(i)} = (\tilde{\mathbf{A}}^{(i)}, \tilde{\mathbf{X}}^{(i)})$, where $i \in \{1, 2\}$.

a) Edge perturbation: A portion of the edges in the similarity graph $\mathcal{G}^{(s)}$ are randomly removed. Specifically, a random mask matrix $\mathbf{M}_e \in \{0, 1\}^{n\times n}$ is first generated, and then the original symmetrically normalized adjacency matrix $\tilde{\mathbf{A}}^{(s)}$ of the

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similarity graph is multiplied by $\mathbf{M}_e$ to obtain a new adjacency matrix $\tilde{\mathbf{A}}^{(i)}$ as follows:

$$\tilde{\mathbf{A}}^{(i)} = \tilde{\mathbf{A}}^{(s)} \circ \mathbf{M}_e \tag{5}$$

where $\circ$ denotes the Hadamard product.

b) Node feature perturbation: To enhance feature diversity, a random mask matrix $\mathbf{M}_n \in \{0,1\}^{n\times f}$ is generated to randomly mask certain elements in the node feature matrix $\mathbf{X}$, resulting in a new node feature matrix $\tilde{\mathbf{X}}^{(i)}$ as follows:

$$\tilde{\mathbf{X}}^{(i)} = \mathbf{X} \circ \mathbf{M}_n. \tag{6}$$

Through the above operations, two augmented graphs, $\mathcal{G}^{(1)}$ and $\mathcal{G}^{(2)}$, are generated based on the similarity graph $\mathcal{G}^{(s)}$. Their node representations are updated in the same way as for the similarity graph, using low-pass filters.

C. Knowledge Distillation

To enable effective information sharing across different graph structures, a dual-level knowledge distillation framework is proposed. It first performs primary alignment distillation between the student and teacher models, then applies a co-guided strategy to iteratively refine both models, forming a more advanced teacher for deeper knowledge transfer.

1) Primary Knowledge Distillation: Up to now, separate graphs are fed into the student and teacher networks, and knowledge is exchanged through embedding alignment. This enhances the student model's performance by transferring informative features from the teacher graph, ultimately improving prediction accuracy on unlabeled nodes.

a) Edge matching: For each node pair in the same batch, the Pearson correlation coefficient [26], [27] is computed between the student embedding graph ($\mathcal{G}_e^{(s)}$) and the teacher embedding graph ($\mathcal{G}_e^{(t)}$), ensuring that the relational structure in the student embeddings aligns with that of the teacher. To enforce this, an edge matching loss is defined as follows:

$$L_{\text{edge}} = \|\text{adj}\left(\mathcal{G}_e^s\right) - \text{adj}\left(\mathcal{G}_e^t\right)\|_2 \tag{7}$$

where adj(.) denotes the operation of calculating the Pearson correlation coefficients between nodes in the graph.

b) Node matching: In addition to structural alignment, individual node embeddings from the teacher also carry valuable information. To transfer this to the student, a node matching loss is defined as follows:

$$L_{\text{node}} = \|\text{adj}\left(\mathcal{G}_e^{\text{st}}\right) - \mathbf{I}\|_2 \tag{8}$$

where $\mathbf{I}$ is the identity matrix.

2) Advanced Knowledge Distillation: Although primary knowledge distillation helps the student learn from the teacher, it may introduce bias and lead to information loss. To address this, a co-guided strategy [28] is adopted to better capture the interaction between teacher and student models. It dynamically adjusts their weights and combines their embeddings to form an advanced teacher, which in turn provides enhanced supervision to both models, improving the student's prediction performance. The process is formalized as follows:

$$\tilde{\mathbf{h}}_i^t, \tilde{\mathbf{h}}_i^s = \text{Co_guided}\left(\mathbf{h}_i^t, \mathbf{h}_i^s\right) \tag{9}$$

where $\tilde{\mathbf{h}}_i^t$ and $\tilde{\mathbf{h}}_i^s$ represent the node embeddings obtained from the student GNN and the teacher GNN for node i, respectively, and Co_guided(.,.) serves as the node feature update function.

Based on the enhanced node embeddings, a more advanced teacher is derived using the following equation:

$$\hat{\mathbf{y}}_i^{\text{st}} = \text{softmax}\left(\mathbf{z}_i^s \circ \mathbf{z}_i^t\right) \tag{10a}$$

$$\mathbf{z}_i^s = \sigma\left(\mathbf{W}_1\tilde{\mathbf{h}}_i^s + \mathbf{b}_1\right) \tag{10b}$$

$$\mathbf{z}_i^t = \sigma\left(\mathbf{W}_2\tilde{\mathbf{h}}_i^t + \mathbf{b}_2\right) \tag{10c}$$

where $\hat{\mathbf{y}}_i^{\text{st}}$ is the probability distribution of the advanced teacher, and σ is the activation function. $\mathbf{W}_1$ and $\mathbf{W}_2$ are the two learnable weight matrices.

The logits of the student nodes ($\mathbf{z}_i^{\text{ps}}$) and the primary teacher nodes ($\mathbf{z}_i^{\text{pt}}$) are normalized to obtain $\hat{\mathbf{y}}_i^s$ and $\hat{\mathbf{y}}_i^t$, respectively. The corresponding loss functions are defined as follows:

$$L_{\text{logits1}} = \frac{1}{n}\sum_{i=1}^{n} \text{KL}\left(\hat{\mathbf{y}}_i^{\text{st}} \| \hat{\mathbf{y}}_i^s\right) \tag{11}$$

$$L_{\text{logits2}} = \frac{1}{n}\sum_{i=1}^{n} \text{KL}\left(\hat{\mathbf{y}}_i^{\text{st}} \| \hat{\mathbf{y}}_i^t\right). \tag{12}$$

This process enables the student to learn from the primary teacher while leveraging both models' outputs via the co-guided strategy, leading to a stronger advanced teacher and better use of multigraph information.

D. Model Training

Given the above focal designs, we train the MDKDI model in an iterative learning manner. To achieve knowledge transfer from the teacher model to the student model, we composite the dual-level knowledge distillation losses into an integrated distillation loss, which is formally defined as follows:

$$L_{\text{KD}} = \lambda\left(L_{\text{node}} + L_{\text{edge}}\right) + \mu\left(L_{\text{logits1}} + L_{\text{logits2}}\right) \tag{13}$$

where λ and μ are hyperparameters balancing the primary and advanced knowledge distillation losses.

To enable end-to-end training of the student model, we jointly optimize the classification loss and knowledge distillation loss. The complete training objective is formally defined as follows:

$$L = L_{\text{main}} + L_{\text{KD}} \tag{14}$$

where L_{main} represents the cross-entropy loss between the model predictions $\hat{\mathbf{Y}}$ and the ground truth labels $\mathbf{Y}$.

IV. Experiments

In this section, we conduct extensive experiments to validate the effectiveness of MDKDI, aiming to address the following research questions.

RQ1: How does MDKDI perform in few-shot scenarios compared to existing baseline methods?

RQ2: What is the contribution of the core components of the MDKDI framework to overall performance?

RQ3: How do different hyperparameter settings affect the performance of MDKDI?

RQ4: How can the performance enhancement achieved by MDKDI during the knowledge distillation process be effectively and intuitively visualized?

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A. Dataset Description

Experiments are conducted on two publicly available DGA datasets: DGA200[1] and DGA2000.[2] These datasets are derived from oil-immersed power transformers in real-world operation and record the concentrations of key dissolved gas components in transformer oil (including H_2, CH_4, C_2H_6, C_2H_4, and C_2H_2) along with corresponding fault category labels.

To better simulate different few-shot classification scenarios, two data splitting methods are employed to evaluate the model's performance.

1) *Proportional splitting:* A random selection of 1%, 2%, 3%, 4%, and 5% of the dataset is used as the training set, while the remaining data constitutes the test set. This setting reflects realistic supervised learning scenarios where the number of samples per class may be imbalanced.
2) *Per-class fixed sample splitting:* A fixed number of samples {1, 2, 3, 4, 5} are randomly selected from each class to form the training set, with the remaining samples used to construct the test set. This setting simulates few-shot learning conditions, ensuring that each class contributes equally to the training set. By adopting both splitting strategies, we can comprehensively evaluate the effectiveness and robustness of MDKDI under different label distribution scenarios.

B. Experimental Settings

1) Evaluation Metrics: The performance is comprehensively evaluated using accuracy (ACC) and $F1$-score ($F1$). The best performance is highlighted in bold, and the second-best performance is underlined. Each experiment is repeated ten times, and the mean and standard deviation of the results are reported in the format mean $\pm$ std.

2) Baseline Methods: We compare against three categories of representative baseline methods: traditional machine-learning paradigms, improved diagnostic methods, and state-of-the-art graph learning approaches:

1) *Traditional Machine-Learning Methods.*
   a) *KNN [11]:* Classifies samples based on the majority class among their KNNs.
   b) *BPNN [19]:* Trains a multilayer neural network using backpropagation to minimize classification error.
2) *Improved Diagnostic Methods.*
   a) *WSO SVM [12]:* Enhances SVM performance by combining feature engineering with whale swarm optimization for parameter tuning.
   b) *IMBP [20]:* Integrates residual networks and SVM into an improved BP neural network for transformer fault diagnosis.
   c) *PIGFD [21]:* Incorporates domain knowledge into the learning process to improve robustness and interpretability in transformer fault diagnosis.
3) *Graph Learning Methods.*
   a) *GCDGA [22]:* Constructs a dual-channel graph encoder and applies contrastive loss to enhance node representations.
   b) *GraphSmin [23]:* Addresses class imbalance in DGA by generating minority class samples using contrastive graph filters.
   c) *DGRCL [24]:* Extends dual-channel graph representation with ranking contrastive learning to achieve more discriminative embeddings.

3) Implementation Details: All experiments are conducted on a server equipped with an Nvidia RTX 4090D GPU, using PyTorch 2.1.2 as the deep learning framework. For the proposed method, the Adam optimizer is employed to optimize network parameters. A learning rate of 0.001 is used for both the DGA200 and DGA2000 datasets. The hidden layer dimensions are set to 128 and 256, and the model depth is set to two and three layers, respectively. The number of KNNs (K) is set to 2 and 16 to effectively aggregate information from neighboring nodes. For the dual-level knowledge distillation strategy, the weight coefficients are set to 0.2 to balance knowledge transfer between the student and teacher networks. For the comparative baselines, KNN was implemented using its standard Python library implementation, while other baselines were reproduced faithfully from their original publications. Unlike the larger scale training sets or partitioning rules used in prior studies, all methods were evaluated under few-shot conditions using both proportional and balanced per-class sample splitting strategies.

C. Performance Comparison (RQ1)

This section compares the performance of the MDKDI model with other baseline methods under the two aforementioned few-shot settings to ensure the comprehensiveness and accuracy of the results.

1) Performance Under Proportional Splitting: Table I reports the performance of the MDKDI and various baseline methods at different sample proportions $\{1\%, 2\%, 3\%, 4\%, 5\%\}$. It is evident that MDKDI exhibits significant advantages in most cases. This superiority is primarily attributed to,

1) The multigraph structure, which captures the complex interactions among samples, effectively alleviating overfitting caused by sample sparsity.
2) The "primary-advanced" dual-level knowledge distillation strategy, which enables diverse information interaction and sharing, providing the student network with more comprehensive guidance. As the sample proportion decreases, the superiority of MDKDI becomes more pronounced, further validating the effectiveness of the multigraph modeling and knowledge distillation mechanism. Interestingly, PIGFD shows an upward trend in performance as the proportion of training samples decreases, achieving the best ACC at 1% on the DGA200 dataset. This may be because PIGFD is highly sensitive to data quality and was specifically designed for scenarios with limited data availability, thus showing an advantage in extremely low-sample conditions. It is worth noting that GraphSmin achieves the best ACC

[1]https://IEEE-dataport.org/documents/dissolved-gas-data-transformer-oil-fault-diagnosis-power-transformers-membership-degree
[2]https://github.com/alan-456/transformer-fault-dataset

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TABLE I
PERFORMANCE COMPARISON BETWEEN MDKDI AND BASELINE METHODS UNDER DIFFERENT PROPORTIONAL SPLITTING

Dataet	Method	5%		4%		3%		2%		1%	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
DGA200	*K*NN	47.83±7.64	45.97±8.43	39.68±5.22	37.97±5.96	41.83±6.80	40.43±8.23	28.60±6.04	25.58±6.11	21.49±4.38	17.45±2.95
	BPNN	49.05±11.47	48.89±10.92	43.32±8.59	43.50±8.77	42.62±11.97	42.75±12.06	35.91±6.78	35.59±7.07	31.03±8.48	29.05±10.25
	WSO SVM	50.21±3.32	42.96±4.64	49.63±5.60	39.76±6.68	49.21±4.36	43.49±4.63	46.27±6.40	42.44±7.91	43.21±9.37	41.34±9.75
	IMBP	55.77±14.01	47.57±16.94	52.05±13.98	41.63±17.68	53.19±7.33	45.16±9.21	46.63±10.68	34.07±11.97	44.79±9.25	39.97±10.63
	PIGFD	56.45±2.96	44.72±2.44	44.01±1.26	34.71±1.74	46.09±4.49	34.12±3.56	48.09±1.32	44.94±1.11	**51.89±4.16**	44.51±4.15
	GCDGA	59.26±5.39	58.11±5.71	55.47±3.64	53.65±3.97	52.93±5.27	51.48±5.67	48.03±5.66	45.83±7.25	45.39±11.89	44.04±12.36
	GraphSmin	59.37±6.25	51.76±4.11	55.53±5.28	47.07±3.99	56.28±6.99	49.10±5.54	51.19±6.82	41.62±7.08	47.58±10.33	40.29±8.99
	DGRCL	58.57±6.18	56.38±6.33	56.84±9.67	53.91±11.24	54.82±3.71	53.32±3.52	54.51±6.17	51.84±7.81	44.12±4.82	35.99±6.71
	$MDKDI_1$	60.58±6.35	59.31±5.94	**59.26±5.39**	**58.12±5.56**	59.79±5.35	59.38±5.48	**55.85±6.65**	**54.26±5.65**	50.36±8.81	**49.22±9.38**
	$MDKDI_2$	**61.32±4.07**	**61.66±3.68**	58.42±3.68	56.82±4.10	**60.05±5.34**	**59.63±5.7**5	54.56±6.62	52.91±5.96	49.43±9.59	48.43±9.64
DGA2000	*K*NN	71.16±2.11	70.20±2.23	70.52±2.65	69.90±2.55	68.23±2.85	67.49±2.61	64.58±3.32	64.28±3.01	58.71±5.37	58.28±5.62
	BPNN	70.30±3.66	69.76±3.84	68.80±4.68	68.38±4.69	67.69±3.89	67.61±3.98	64.73±4.96	64.39±4.79	61.88±5.58	62.50±4.87
	WSO SVM	71.75±1.28	68.01±1.16	65.68±1.31	59.19±1.73	62.75±1.73	55.29±2.59	63.35±1.93	56.05±2.49	64.09±2.65	57.84±3.22
	IMBP	65.61±8.30	57.31±11.73	67.74±9.48	60.59±12.9	62.94±7.36	54.59±10.17	67.38±8.22	59.65±11.91	66.21±6.39	59.05±8.99
	PIGFD	64.77±2.03	40.92±1.45	65.37±0.90	45.80±1.20	66.63±1.17	46.03±1.29	62.81±1.01	42.80±1.59	60.91±0.94	41.18±1.42
	GCDGA	70.47±1.08	69.11±1.43	69.95±1.78	68.33±1.73	67.46±2.93	66.68±3.07	67.79±2.70	67.21±2.76	64.24±2.50	63.56±2.40
	GraphSmin	**75.48±2.26**	65.44±2.32	**75.48±1.87**	65.26±1.98	72.55±2.56	61.94±2.92	71.02±2.33	60.91±3.30	69.28±2.10	58.76±2.19
	DGRCL	71.33±2.12	71.17±1.87	69.91±2.29	70.20±2.46	68.36±2.75	69.24±2.25	67.44±1.49	68.23±1.52	65.42±2.05	65.85±1.86
	$MDKDI_1$	75.12±1.05	**73.96±0.92**	73.66±1.39	**72.92±1.37**	**73.15±1.75**	**72.21±2.28**	**71.66±3.08**	**71.24±2.77**	**70.41±2.40**	**69.33±2.51**
	$MDKDI_2$	74.38±0.73	72.45±0.72	73.04±1.54	71.81±1.58	72.88±2.60	70.27±3.37	71.24±2.51	69.28±2.05	69.36±1.91	68.49±1.17

TABLE II
PERFORMANCE COMPARISON BETWEEN MDKDI AND BASELINE METHODS UNDER FIXED PER-CLASS SAMPLE SPLITTING

Dataet	Method	5		4		3		2		1	
		ACC	F1	ACC	F1	ACC	F1	ACC	F1	ACC	F1
DGA200	*K*NN	52.29±4.43	53.23±5.37	48.73±6.39	50.06±7.10	43.44±5.53	41.04±8.79	40.75±5.14	37.23±7.28	26.24±6.03	19.69±3.28
	BPNN	55.36±10.33	56.18±9.40	50.22±15.80	49.86±15.25	50.22±15.80	49.86±15.25	43.74±12.03	44.26±12.02	37.58±10.41	7.05±4.36
	WSO SVM	51.69±5.33	52.86±4.54	49.94±2.66	51.15±2.77	46.72±4.43	45.92±5.36	52.67±4.80	51.95±5.01	41.02±6.64	40.49±7.86
	IMBP	63.80±8.00	59.32±12.41	56.76±18.69	54.19±20.07	54.39±14.77	51.99±15.65	55.35±12.27	51.87±14.54	44.18±12.75	39.14±12.84
	PIGFD	50.29±0.96	45.95±1.70	50.12±1.90	42.17±2.92	46.19±1.90	37.64±3.39	51.11±3.20	46.48±2.40	45.60±3.46	37.54±4.12
	GCDGA	64.04±4.89	64.18±4.75	62.43±4.42	62.67±4.66	59.50±2.74	59.05±3.34	58.29±6.53	58.01±6.15	45.41±6.53	45.07±5.98
	GraphSmin	64.34±5.35	53.72±5.01	65.20±4.42	55.63±4.27	60.44±4.10	50.58±2.88	60.32±4.61	51.91±3.57	46.70±7.99	40.72±5.95
	DGRCL	65.30±4.65	65.69±3.96	64.34±4.82	64.23±4.84	**64.06±4.21**	63.06±5.54	59.52±6.60	57.64±8.30	**53.81±6.65**	**51.59±6.45**
	$MDKDI_1$	67.89±4.03	67.48±3.86	65.03±3.13	65.16±3.33	63.17±3.57	62.72±3.70	**63.80±2.99**	**63.63±2.70**	51.03±9.75	49.71±9.49
	$MDKDI_2$	**68.31±4.09**	**68.22±4.11**	**66.01±2.85**	**66.15±3.48**	63.94±4.34	**64.71±4.71**	62.99±3.92	63.07±3.49	50.05±9.02	49.90±9.36
DGA2000	*K*NN	57.41±4.10	58.92±4.52	55.80±4.18	56.54±4.68	51.53±5.13	51.23±6.88	48.86±4.75	46.95±5.32	38.82±9.92	33.38±9.91
	BPNN	61.86±7.04	63.76±7.27	57.61±4.94	58.84±5.67	54.80±5.71	56.80±5.05	53.88±9.15	55.68±8.94	40.11±8.68	40.74±9.09
	WSO SVM	60.25±3.59	62.02±3.41	56.35±2.28	57.18±2.44	59.27±5.38	60.76±5.16	47.78±6.24	47.04±6.28	46.96±10.46	47.03±13.73
	IMBP	59.22±8.87	58.88±9.30	60.77±9.89	60.06±11.03	58.44±5.02	58.87±6.40	47.32±13.53	40.42±17.11	46.19±10.19	41.04±11.42
	PIGFD	58.13±1.08	46.03±1.62	60.29±1.39	51.18±1.77	58.51±1.32	47.15±1.83	59.51±2.51	47.64±1.47	53.18±1.49	42.26±1.83
	GCDGA	64.25±2.68	64.90±2.55	61.30±3.99	61.72±3.27	60.80±3.68	61.22±3.64	52.90±3.97	53.97±3.94	50.16±5.21	50.43±4.93
	GraphSmin	**68.71±5.00**	60.57±5.09	**67.95±4.05**	59.64±3.97	64.66±4.37	56.54±3.06	56.85±7.77	50.54±6.03	51.14±12.09	45.15±7.12
	DGRCL	66.36±2.29	67.99±2.37	65.60±3.03	66.97±3.13	65.01±4.42	65.80±3.92	58.02±4.45	59.41±4.16	52.68±4.77	53.66±3.79
	$MDKDI_1$	68.24±3.98	**69.54±3.63**	67.92±3.57	**69.16±3.28**	**66.75±5.41**	**68.15±4.54**	**59.74±4.80**	**61.53±4.87**	**58.34±4.39**	**58.82±4.06**
	$MDKDI_2$	67.87±3.98	68.79±3.62	67.20±3.84	67.95±2.85	65.37±3.63	66.68±3.36	58.44±4.65	59.88±4.08	58.34±4.56	58.33±4.57

results for {4%, 5%} on the DGA2000 dataset, demonstrating that its strategy of generating minority class samples using contrastive dual-channel graph filters is particularly effective in certain scenarios.

2) Performance Under Fixed Per-Class Sample Splitting: To further evaluate the performance of MDKDI, a balanced per-class sampling strategy is adopted for training, where $\{1, 2, 3, 4, 5\}$ samples are selected from each class to construct the training set. The results are shown in Table II. MDKDI still demonstrates outstanding performance in most cases, especially under the extreme condition of only one sample per class, where it achieves a 5% improvement in $F1$ on the DGA2000 dataset compared to the best baseline DGRCL.

Combining the results from Table I, the following observations can be made.

1) On the DGA200 dataset, DGRCL achieves the best results in terms of 4-shot ACC as well as 1-shot ACC and $F1$, indicating that its ranking contrastive learning design provides more discriminative embeddings under extremely few-shot conditions. On the DGA2000 dataset, GraphSmin achieves the best ACC for {4, 5} samples, further demonstrating the effectiveness of generating minority class samples through contrastive graph filters for transformer fault diagnosis.
2) Despite the competitiveness of these strong baselines, the proposed MDKDI consistently demonstrates superior performance across both proportional and fixed per-class splitting strategies, confirming the robustness and general effectiveness of the multigraph modeling and dual-level knowledge distillation mechanism in practical few-shot DGA scenarios.

Overall, the consistent superiority of MDKDI under both proportional and per-class fixed sample splitting indicates that the model is robust to fundamentally different data distributions. This robustness mainly stems from the multigraph construction, which mitigates the bias of imbalanced or scarce data, and the dual-level knowledge distillation strategy, which provides complementary guidance to enhance generalization.

3) Efficiency Analysis: To further assess the efficiency of the proposed MDKDI framework, we evaluate it from both theoretical and empirical perspectives. Theoretically, the computational cost of MDKDI mainly arises from three components: graph filtering, knowledge distillation, and col-

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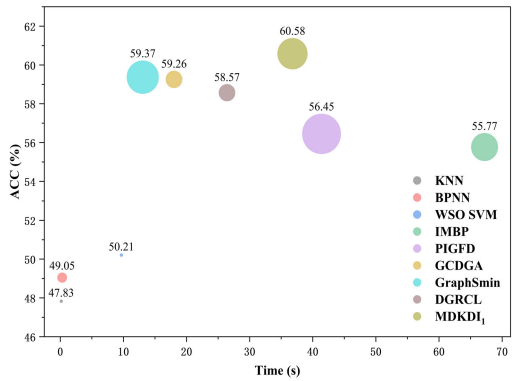


Fig. 2. Efficiency analysis of $MDKDI_1$ on the DGA200 dataset with 5% training ratio. The horizontal axis denotes the training runtime of each model, the vertical axis indicates the accuracy (ACC), and the bubble size represents the cube root of the number of trainable parameters.

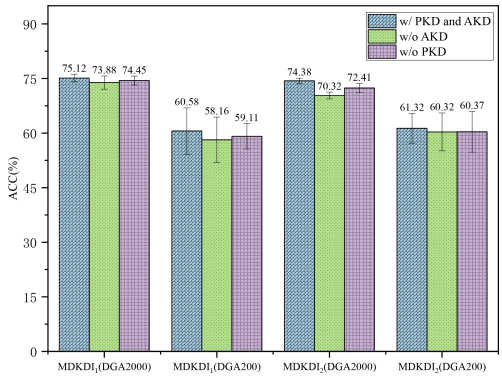


Fig. 3. Effectiveness analysis of knowledge distillation.

laborative guidance. Graph filtering has a time complexity of $O(qf+qd)$, where q denotes the number of nonzero elements in the adjacency matrix, f is the input feature dimension, and d is the hidden dimension. Knowledge distillation involves global similarity computations, with a complexity of $O(md)$, where m is the number of edges in the graph. In the collaborative guidance module, the model refines embeddings through bilinear interactions for information fusion, with a time complexity of $O(nd^2)$, where n denotes the number of samples. Therefore, the overall time complexity of MDKDI can be expressed as $O(qf+qd+md+nd^2)$. It is worth noting that KNN graph construction is a one-time preprocessing step and does not affect the training phase efficiency.

As shown in Fig. 2, under a 5% training ratio on the DGA200 dataset, traditional methods (KNN, BPNN) exhibit the lowest computational cost but also the lowest accuracy ($<$ 50%), rendering them inadequate for complex tasks. Improved methods (WSO SVM, IMBP, PIGFD) yield moderate accuracy gains at the expense of higher complexity, with PIGFD peaking at 56.45% ACC. In contrast, graph-based approaches demonstrate superior performance: GraphSmin (59.37%) and GCDGA (59.26%) achieve high accuracy, with GCDGA maintaining low computational cost. Our proposed $MDKDI_1$ strikes the optimal balance, achieving the highest accuracy (60.58%) while retaining reasonable runtime and parameter efficiency.

D. Component Effectiveness Analysis (RQ2)

This section evaluates the effectiveness of the key components of the MDKDI model, focusing on the rationale behind the KNN graph construction and the effectiveness of the knowledge distillation strategy.

1) Effectiveness of Knowledge Distillation: To thoroughly validate the effectiveness of the "primary-advanced" knowledge distillation architecture, extensive experiments are conducted on the DGA200 and DGA2000 datasets. The experiments are equipped with the following model configurations.

Full model (w/ PKD and AKD): Incorporates both primary knowledge distillation (PKD) and advanced knowledge distillation (AKD).

TABLE III
RATIONALITY VERIFICATION FOR KNN GRAPH CONSTRUCTION

ϵ	DGA200		DGA2000	
	ACC	F1	ACC	F1
0.3	52.22±4.45	48.17±3.85	48.00±2.91	40.83±4.31
0.5	53.81±4.43	50.68±3.55	64.19±2.48	60.18±3.67
0.7	55.24±5.52	53.85±4.90	70.00±1.38	67.34±1.44
0.9	57.72±3.65	56.74±3.48	71.33±0.65	67.9±0.84
$MDKDI_1$	**60.58±6.35**	**59.31±5.94**	**75.12±1.05**	**73.96±0.92**

Without primary knowledge distillation (without PKD): Retains only AKD.

Without advanced knowledge distillation (without AKD): Retains only PKD.

The experimental results are shown in Fig. 3, and we have the following observations.

The full model achieves the best performance on both the datasets, clearly demonstrating that the combination of PKD and AKD provides complementary information, thereby improving overall prediction accuracy. When the PKD loss is removed, the model performance decreases to varying degrees. This phenomenon confirms that PKD loss brings rich structural information to the model through node matching and edge matching constraints. In particular, under sparse data conditions, PKD loss effectively suppresses the propagation of topological noise, ensuring learning quality. On the other hand, when AKD loss is removed, the model performance also falls short of the full model. This underlines the critical role of AKD loss in the knowledge distillation process, where the advanced teacher provides deeper guidance to the student network and enhances the model's capacity. By adopting the dual-level knowledge distillation strategy, the model not only improves learning efficiency but also significantly enhances generalization ability. This strategy enables the model to better capture and utilize the complex relationships in the data, demonstrating robust performance across multiple tasks. Moreover, the experiments reveal the dual-level distillation architecture's adaptability to different datasets. Whether on the smaller DGA200 dataset or the larger DGA2000 dataset, the full model consistently achieves outstanding results, indicating that MDKDI exhibits excellent adaptability.

2) Rationale of KNN Graph Construction: To verify the validity of the KNN graph construction strategy, this section compares the proposed KNN method with the ϵ-neighborhood

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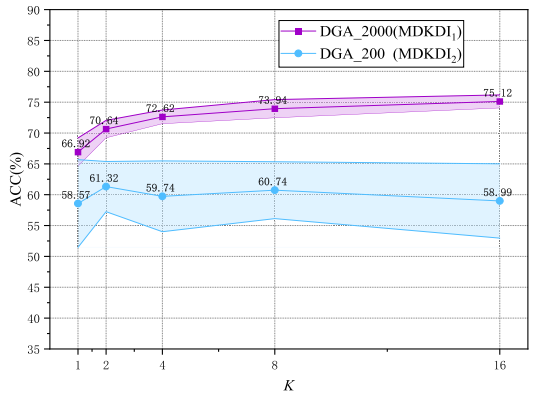


Fig. 4. ACC values of MDKDI under different K values.

graph under a 5% training set proportion. In this experiment, the number of KNNs is set to the optimal value determined in Section IV-E1 and $\epsilon \in \{0.3, 0.5, 0.7, 0.9\}$. The experimental results are shown in Table III.

The results show that MDKDI_1 consistently achieves the best performance, strongly supporting the rationale of the KNN graph construction strategy. The possible reasons for this are analyzed as follows: 1) KNN can adaptively adjust the local neighborhood structure, while the ϵ-neighborhood graph is more sensitive to density variations and 2) KNN reduces the influence of noise by selecting the nearest K points, whereas the ϵ-neighborhood graph may include noise points within the neighborhood. Furthermore, given the larger scale of the DGA2000 dataset, constructing an ϵ-neighborhood graph incurs a relatively high computational cost, whereas the KNN method demonstrates significant advantages in computational efficiency. Therefore, the KNN graph construction strategy not only performs well in terms of accuracy but is also more efficient in practical applications, making it particularly suitable for handling larger scale datasets.

E. Hyperparameter Analysis (RQ3)

This section investigates the impact of the number of neighbors K in the KNN graph and the knowledge distillation weight parameters on the model's performance, aiming to verify the sensitivity of the MDKDI framework to hyperparameters.

1) Analysis of K: Under the 5% training set configuration, the performance of MDKDI is evaluated on the DGA200 and DGA2000 datasets for $K \in \{1, 2, 4, 8, 16\}$ (as shown in Fig. 4). The experimental findings reveal the following insights:

For the DGA2000 dataset, MDKDI_1 exhibits strong stability in performance. Notably, when $K = 1$, the model performance drops significantly (ACC = 66.92%, 8% points lower than the optimal value). This phenomenon can be attributed to: 1) when $K = 1$, the information propagation among unlabeled samples is restricted, leading to insufficient information utilization and 2) single-neighbor selection is prone to being influenced by outliers, making the model more sensitive to noise. Based on the above analysis, $K = 16$ is selected as the optimal hyperparameter for the DGA2000 dataset, achieving a balance between information utilization and noise robustness.

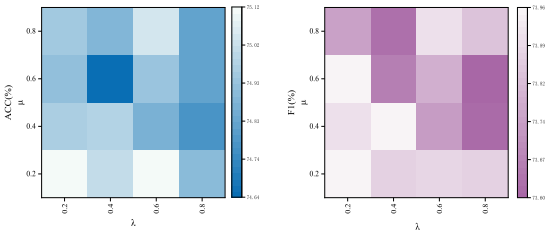


Fig. 5. Knowledge distillation weight analysis of MDKDI_1 on the DGA2000 dataset.

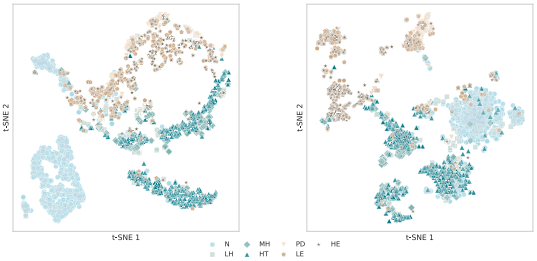


Fig. 6. Comparative visualization of embedded features on the DGA2000 dataset.

For the DGA200 dataset, MDKDI_2 achieves the best results when $K = 2$. When $K = 1$, performance decreases by approximately 3% for reasons similar to those observed with the DGA2000 dataset. Conversely, when K is too large, performance degrades significantly. This can be explained as follows: 1) in smaller scale datasets, an excessively large K leads to over-smoothing of the network topology and 2) as K increases, the likelihood of including incorrect connections also rises. Therefore, $K = 2$ is selected as the optimal hyperparameter for the DGA200 dataset, achieving a balance between information aggregation and preserving the authenticity of the topology structure.

2) Analysis of Knowledge Distillation Weights: To evaluate the weight allocation mechanism of dual-level knowledge distillation, a grid search strategy is employed to explore the impact of hyperparameters λ (PKD weight) and μ (AKD weight) on MDKDI_1's performance on the DGA2000 dataset. The search range is set to $\lambda, \mu \in \{0.2, 0.4, 0.6, 0.8\}$, and the experimental results are shown in Fig. 5. The results indicate that the changes in ACC and $F1$ scores are relatively stable, demonstrating that the model is robust to variations in λ and μ. This stability can be attributed to the complementary functions of primary and advanced distillation: the former focuses on local structural alignment, while the latter focuses on global semantic matching.

Although overall performance fluctuations are small, the model achieves optimal performance (ACC = 75.12%, $F1$ = 73.96%) when $\lambda = 0.2$ and $\mu = 0.2$. Therefore, $\lambda = 0.2$ and $\mu = 0.2$ are chosen to optimize the model's performance.

F. Case Studies (RQ4)

To demonstrate the effectiveness of knowledge distillation, the well-trained MDKDI1 model is employed, and the last hidden representations before the classification head are extracted as embeddings. On the DGA2000 test set with 5% training data, each sample is represented by a 256-dimensional

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embedding vector. The t-SNE algorithm [29] is then applied to reduce both the original features (Fig. 6, left) and the learned embeddings (Fig. 6, right) to two dimensions for visualization. The comparison shows that the embeddings learned by MDKDI_1 achieve more distinct class separation for **PD**, **HE**, and **LE**, indicating that knowledge distillation reduces noise and redundant information, thereby improving embedding quality. However, **MH** and **LH** still lack distinct clusters, which may be due to: 1) the original features of these classes lacking sufficient discriminative information, making them hard to separate even after distillation or 2) the loss of subtle but important distinctions during the distillation process, hindering effective separation in the embedding space.

Overall, knowledge distillation significantly improves feature learning and class discrimination, particularly for **PD**, **HE**, and **LE**, while leaving room for improvement in **MH** and **LH** separation.

V. Conclusion

This article introduces the MDKDI method, designed for few-shot DGA fault diagnosis, to tackle the performance limitations of DGA in scenarios with limited samples. By constructing multigraph structures and employing a dual-level knowledge distillation framework, MDKDI effectively integrates multiview information. Experimental results on two public datasets demonstrate its strong performance and superior generalization ability, especially in extremely few-shot settings.

While the dual-level distillation mechanism mitigates efficiency concerns, the computational complexity of graph-based methods remains a challenge. Future work will focus on developing more efficient distillation strategies and lightweight graph models to improve practicality. Furthermore, the general design of MDKDI makes it adaptable to other graph-based few-shot learning tasks beyond DGA, highlighting its broader utility and application potential.

References

[1] M. Patil, A. Paramane, S. Das, U. M. Rao, and P. Rozga, "Hybrid algorithm for dynamic fault prediction of HVDC converter transformer using DGA data," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 31, no. 4, pp. 2128–2135, Aug. 2024.

[2] M. R. Hussain, S. S. Refaat, and H. Abu-Rub, "Overview and partial discharge analysis of power transformers: A literature review," *IEEE Access*, vol. 9, pp. 64587–64605, 2021.

[3] H. C. Chen and Y. Zhang, "Rethinking shallow and deep learnings for transformer dissolved gas analysis: A review," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 32, no. 1, pp. 3–10, Feb. 2025.

[4] M. S. Ali, A. H. Abu Bakar, A. Omar, A. S. Abdul Jaafar, and S. H. Mohamed, "Conventional methods of dissolved gas analysis using oil-immersed power transformer for fault diagnosis: A review," *Electr. Power Syst. Res.*, vol. 216, Mar. 2023, Art. no. 109064.

[5] A. Akbari, A. Setayeshmehr, H. Borsi, and E. Gockenbach, "A software implementation of the duval triangle method," in *Proc. Conf. Rec. IEEE Int. Symp. Electr. Insul.*, Jun. 2008, pp. 124–127.

[6] H. Malik and S. Mishra, "Application of gene expression programming (GEP) in power transformers fault diagnosis using DGA," *IEEE Trans. Ind. Appl.*, vol. 52, no. 6, pp. 4556–4565, Nov. 2016.

[7] N. Pattanadech and W. Wattakapaiboon, "Application of duval pentagon compared with other DGA interpretation techniques: Case studies for actual transformer inspections including experience from power plants in Thailand," in *Proc. 5th Int. Conf. Eng., Appl. Sci. Technol. (ICEAST)*, Jul. 2019, pp. 1–4.

[8] M. Meira, I. Carlucho, R. Álvarez, L. Catalano, and G. Acosta, "DGA: A novel strategy for key gases identification in power transformers," in *Proc. IEEE Electr. Insul. Conf. (EIC)*, Jun. 2020, pp. 290–293.

[9] L. Hong, Z. Chen, Y. Wang, M. Shahidehpour, and M. Wu, "A novel SVM-based decision framework considering feature distribution for power transformer fault diagnosis," *Energy Rep.*, vol. 8, pp. 9392–9401, Nov. 2022.

[10] O. Kherif, Y. Benmahamed, M. Teguar, A. Boubakeur, and S. S. M. Ghoneim, "Accuracy improvement of power transformer faults diagnostic using KNN classifier with decision tree principle," *IEEE Access*, vol. 9, pp. 81693–81701, 2021.

[11] M. A. N. D. Sewwandi, Y. Li, and J. Zhang, "A class-specific feature selection and classification approach using neighborhood rough set and K-nearest neighbor theories," *Appl. Soft Comput.*, vol. 143, Aug. 2023, Art. no. 110366.

[12] K. Li, K. Wang, G. Yang, B. Lu, J. Jia, and T. Sun, "Fault diagnosis method of transformer based on SWO and SVM," in *Proc. 4th Power Syst. Green Energy Conf. (PSGEC)*, Aug. 2024, pp. 926–930.

[13] Y. Mahamdi, A. Mekhaldi, A. Boubakeur, and Y. Benmahamed, "Advanced SVM and KNN algorithms for fault detection in power transformers based on dissolved gas analysis," *IEEE Trans. Dielectr. Electr. Insul.*, early access, 2025.

[14] S. A. M. Abdelwahab, I. B. M. Taha, R. Fahim, and S. S. M. Ghoneim, "Transformer fault diagnose intelligent system based on DGA methods," *Sci. Rep.*, vol. 15, no. 1, p. 8263, Mar. 2025.

[15] H. C. Chen, Y. Zhang, and M. Chen, "Transformer dissolved gas analysis for highly-imbalanced dataset using multiclass sequential ensembled ELM," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 30, no. 5, pp. 2353–2361, Oct. 2023.

[16] A. Jaysing Patil, R. Naresh, R. Kumar Jarial, and H. Malik, "Optimized synthetic data integration with transformer's DGA data for improved ML-based fault identification," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 32, no. 1, pp. 598–607, Feb. 2025.

[17] L. Wang, T. Littler, and X. Liu, "Gaussian process multi-class classification for transformer fault diagnosis using dissolved gas analysis," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 28, no. 5, pp. 1703–1712, Oct. 2021.

[18] X. Shi, T. Li, F. Fang, Y. Zhu, W. Yang, and B. Luo, "Dissolved gas analysis for power transformer fault diagnosis combining domain knowledge and capsule network," *IEEE Trans. Dielectr. Electr. Insul.*, vol. 31, no. 6, pp. 3386–3395, Dec. 2024.

[19] J. Yuan, P. Xu, L. Li, Y. Liu, X. Wang, and Y. Zheng, "Prediction of transformer oilpaper insulation aging based on BP neural networks with the chicken swarm optimization algorithm," *J. Electr. Power Sci. Technol.*, vol. 35, no. 4, pp. 33–41, 2020.

[20] Y. Jin, H. Wu, J. Zheng, J. Zhang, and Z. Liu, "Power transformer fault diagnosis based on improved BP neural network," *Electronics*, vol. 12, no. 16, p. 3526, Aug. 2023.

[21] L. Lei, Y. He, and Z. Xing, "Physics-informed guided transformer fault diagnosis via DGA data," *IEEE Trans. Dielectr. Electr. Insul.*, early access, 2025.

[22] Y. Zhang, H. Ma, D. Zhang, K. Shu, and X. Li, "Graph contrastive learning for dissolved gas analysis," in *Proc. Int. Conf. Adv. Data Mining Appl.*Cham, Switzerland: Springer, 2024, pp. 178–190.

[23] K. Shu, H. Ma, J. Yang, and D. Zhang, "GraphSmin: Imbalanced dissolved gas analysis with contrastive dual-channel graph filters," *Adv. Eng. Informat.*, vol. 62, Oct. 2024, Art. no. 102839.

[24] Y. Zhang, H. Ma, D. Zhang, K. Shu, and X. Li, "Dual-channel graph ranking contrastive learning for dissolved gas analysis," *IEEE Trans. Ind. Informat.*, vol. 21, no. 8, pp. 5933–5942, Aug. 2025.

[25] W. Jin, H. Ma, Y. Zhang, Z. Li, and L. Chang, "Multi-view discriminative edge heterophily contrastive learning network for attributed graph anomaly detection," *Expert Syst. Appl.*, vol. 255, Dec. 2024, Art. no. 124460.

[26] Y. Ma, Y. Chen, and Z. Akata, "Distilling knowledge from self-supervised teacher by embedding graph alignment," 2022, *arXiv:2211.13264*.

[27] I. Cohen et al., "Pearson correlation coefficient," in *Noise Reduction in Speech Processing*. Berlin, Germany: Springer, 2009, pp. 1–4.

[28] Z. Wu, H. Ma, B. Deng, Z. Li, and L. Chang, "Co-guided dual-channel graph neural networks for the prediction of compound–protein interaction," *Appl. Soft Comput.*, vol. 163, Sep. 2024, Art. no. 111875.

[29] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

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